文献阅读笔记

自旋转桨轮两栖机器人

文献信息 Chaewon Kim, Kyungwook Lee, Sijun Ryu, TaeWon Seo. “Amphibious Robot With Self-Rotating Paddle-Wheel Mechanism.” *IEEE/ASME Transactions on Mechatronics*, 28(4), 2023, 1836–1843. DOI: 10.1109/TMECH.2023.3273240。

一句话概括 **论文把“越障能力强的斜辐条轮”改造成会自转的桨轮 ASPW，使桨叶在轮体完全浸没时仍能形成方向不对称的水动力，从而兼顾陆地、浅水、岸滩过渡与障碍跨越。**

研究问题

海事事故现场可能有狭窄通道、坠落物和水陆交界，救援人员与常规船只难以进入。已有两栖机器人往往在水面速度、地面机动、越障、体积和传感器视野之间顾此失彼。作者要解决的是：能否仅用一套紧凑轮系，同时实现陆地行驶、水面推进和较强越障？

核心机制

ASPW 由 6 个桨叶和两级齿轮组构成。轮毂转动时，行星齿轮与外齿轮约束桨叶同步自转，使桨叶迎水投影面积随相位周期变化：上方回程阶段投影面积最小，反向阻力小；下方做功阶段投影面积最大，正向推力大。净推进力因此不再互相抵消。

齿轮设计的关键点是：外齿轮组采用 1:1:2 关系，使顶部与底部桨叶互相垂直；太阳轮 48 齿、行星轮 12 齿，使 6 个桨叶保持稳定的 30°相位间隔。主体金属件采用不锈钢/黄铜，桨叶用 ASA 材料 3D 打印。

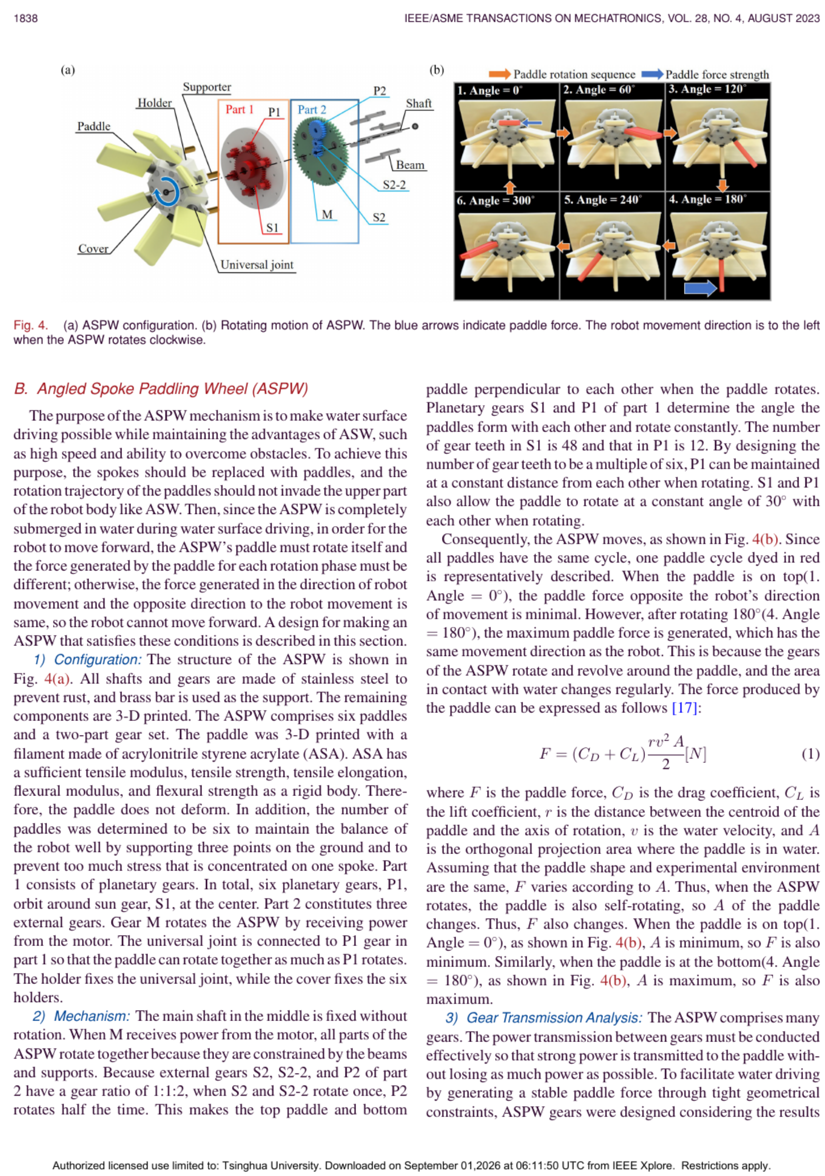

Fig. 4. (a) ASPW configuration. (b) Rotating motion of ASPW. The blue arrows indicate paddle force. The robot movement direction is to the left when the ASPW rotates clockwise.

B. Angled Spoke Paddling Wheel (ASPW)

The purpose of the ASPW mechanism is to make water surface driving possible while maintaining the advantages of ASW, such as high speed and ability to overcome obstacles. To achieve this purpose, the spokes should be replaced with paddles, and the rotation trajectory of the paddles should not invade the upper part of the robot body like ASW. Then, since the ASPW is completely submerged in water during water surface driving, in order for the robot to move forward, the ASPW's paddle must rotate itself and the force generated by the paddle for each rotation phase must be different; otherwise, the force generated in the direction of robot movement and the opposite direction to the robot movement is same, so the robot cannot move forward. A design for making an ASPW that satisfies these conditions is described in this section.

1) Configuration: The structure of the ASPW is shown in Fig. 4(a). All shafts and gears are made of stainless steel to prevent rust, and brass bar is used as the support. The remaining components are 3-D printed. The ASPW comprises six paddles and a two-part gear set. The paddle was 3-D printed with a filament made of acrylonitrile styrene acrylate (ASA). ASA has a sufficient tensile modulus, tensile strength, tensile elongation, flexural modulus, and flexural strength as a rigid body. Therefore, the paddle does not deform. In addition, the number of paddles was determined to be six to maintain the balance of the robot well by supporting three points on the ground and to prevent too much stress that is concentrated on one spoke. Part 1 consists of planetary gears. In total, six planetary gears, P1, orbit around sun gear, S1, at the center. Part 2 constitutes three external gears. Gear M rotates the ASPW by receiving power from the motor. The universal joint is connected to P1 gear in part 1 so that the paddle can rotate together as much as P1 rotates. The holder fixes the universal joint, while the cover fixes the six holders.

2) Mechanism: The main shaft in the middle is fixed without rotation. When M receives power from the motor, all parts of the ASPW rotate together because they are constrained by the beams and supports. Because external gears S2, S2-2, and P2 of part 2 have a gear ratio of 1:1:2, when S2 and S2-2 rotate once, P2 rotates half the time. This makes the top paddle and bottom paddle perpendicular to each other when the paddle rotates. Planetary gears S1 and P1 of part 1 determine the angle the paddles form with each other and rotate constantly. The number of gear teeth in S1 is 48 and that in P1 is 12. By designing the number of gear teeth to be a multiple of six, P1 can be maintained at a constant distance from each other when rotating. S1 and P1 also allow the paddle to rotate at a constant angle of 30° with each other when rotating.

Consequently, the ASPW moves, as shown in Fig. 4(b). Since all paddles have the same cycle, one paddle cycle dyed in red is representatively described. When the paddle is on top(1. Angle $= 0°$), the paddle force opposite the robot's direction of movement is minimal. However, after rotating 180°(4. Angle $= 180°$), the maximum paddle force is generated, which has the same movement direction as the robot. This is because the gears of the ASPW rotate and revolve around the paddle, and the area in contact with water changes regularly. The force produced by the paddle can be expressed as follows [17]:

$$F = (C_D + C_L)\frac{rv^2 A}{2}[N] \tag{1}$$

where F is the paddle force, C_D is the drag coefficient, C_L is the lift coefficient, r is the distance between the centroid of the paddle and the axis of rotation, v is the water velocity, and A is the orthogonal projection area where the paddle is in water. Assuming that the paddle shape and experimental environment are the same, F varies according to A. Thus, when the ASPW rotates, the paddle is also self-rotating, so A of the paddle changes. Thus, F also changes. When the paddle is on top(1. Angle $= 0°$), as shown in Fig. 4(b), A is minimum, so F is also minimum. Similarly, when the paddle is at the bottom(4. Angle $= 180°$), as shown in Fig. 4(b), A is maximum, so F is also maximum.

3) Gear Transmission Analysis: The ASPW comprises many gears. The power transmission between gears must be conducted effectively so that strong power is transmitted to the paddle without losing as much power as possible. To facilitate water driving by generating a stable paddle force through tight geometrical constraints, ASPW gears were designed considering the results

图：原文第 3 页，展示 ASPW 构型、齿轮约束和桨叶一周内的自转过程。

桨叶优化

由于新机构的流体作用呈强周期性，作者认为高成本数值仿真并不划算，转而采用田口法 L9(3^4) 做实验优化。四个设计变量是 X/Y 方向曲率、桨叶端部形状和安装倾角，外部条件是 60/90/120 rpm。

最优组合为：X 向曲率半径 37.5 mm、Y 向曲率半径 25 mm、矩形端部、0°倾角。在 120 rpm 下，推力比基础桨形提升 18.27%。这部分的价值不只在参数本身，更在于展示了“难建模新机构”可用稳健试验设计快速收敛。

实验与主要结果

工况	结果
平地速度	60/90/120/135 rpm 时平均为 0.19/0.31/0.40/0.47 $m\cdot s^{-1}$；与椭圆轮迹模型预测一致
水面速度	60/90/120/135 rpm 时为 0.025/0.049/0.077/0.10 $m\cdot s^{-1}$
普通桨轮对照	在相同完全浸没条件下无法产生前进位移，支持“自转造成净推力”的机理解释
陆→水过渡	最大可通过约 25°坡面
水→陆过渡	最大可通过约 20°坡面
越障	理论 41.8 mm，实测 41.3 mm，二者接近
野外验证	草地、沙地、低矮空间、碎石地与有流动水的溪流均完成行驶

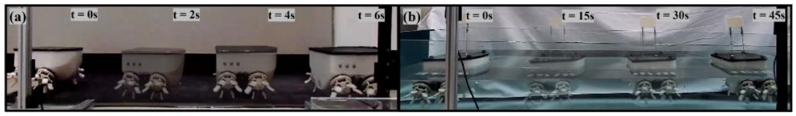

Fig. 8. (a) Ground driving ability experiment. (b) Water surface driving ability experiment. The results of water surface driving experiments of ASPW and conventional paddles were compared on this test bench. Refer to multimedia extension for the experiment.

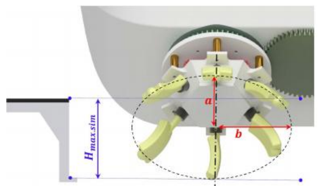

Fig. 9. ASPW trajectory. The short axis of the ellipse is a and the long axis of the ellipse is b. $H_{\max.\mathrm{sim}}$ is the theoretical maximum height that can overcome obstacles.

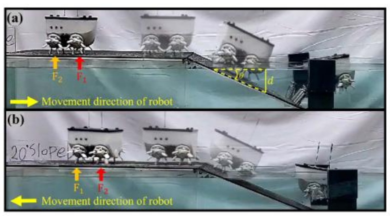

Fig. 10. (a) Ground to water surface transition experiment. (b) Water surface to ground transition experiment. Refer to multimedia extension for the experiment.

on the spoke length, and rotational transformation and projection from the spoke trajectory were performed to predict the height [31]. Prior to the experiment, the maximum obstacle height that the ASPW could overcome was predicted by considering the paddle length of the ASPW in this way. As a result, as shown in Fig. 9, $H_{\max.\mathrm{sim}}$ was 41.8 mm. The experiment confirmed that the robot can overcome obstacles that have a height of 41.3 mm. Although there is a slight difference between the numerical calculation and the experimental results in the maximum height of obstacles that can be overcome, their values have nearly the same ability to overcome obstacles. Fig. 11 describes the robot overcoming a 41.3 mm high obstacle.

Fig. 12 is a graph comparing the percentage of maximum obstacle overcoming height relative to the wheel size of amphibious robots. Robots that performed obstacle overcoming experiments were compared, and it was assumed that there was

Fig. 11. Obstacle overcoming experiment. Refer to multimedia extension for the experiment.

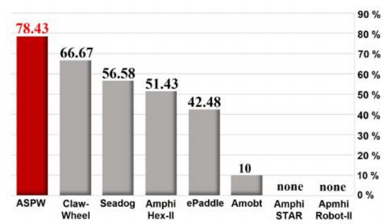

Fig. 12. Percentage of maximum obstacle overcoming height relative to wheel size of amphibious robots.

no meaningful obstacle overcoming height for robots that only performed rough terrain driving experiments. Through Fig. 12, it is possible to obtain insight that the ASPW has an improved obstacle overcoming ability than the existing amphibious robot's wheel mechanism.

F. Field Test

The robot also showed good mobility in various environments. The field test was conducted in a total of five environments: grass field, sand field, low space, gravel field, and stream. Fig. 13(a) describes a driving test on a lawn composed of soft soil. It was confirmed that the spokes can be driven even in uneven environments where the spokes are not rigid. Steering was also possible smoothly, so it was easy to look around. Fig. 13(b) describes a driving test conducted on a sandy playground often studded with large stones. It was confirmed that irregularly shaped obstacles of 41.3 mm or less could be overcome. Fig. 13(c) is an image of exploring under a car assuming a situation where it is difficult for people to enter at a disaster site. The camera mounted on the robot made it possible to grasp the situation instead of a person in an area where it is difficult or dangerous for rescuers to enter. Fig. 13(d) represents a driving test conducted on a gravel field.

图：原文第 7 页，集中给出陆地/水面、过渡坡面和 41.3 mm 障碍实验。

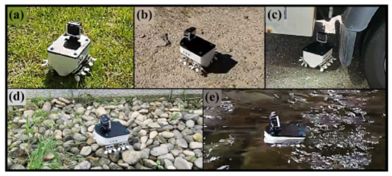

Fig. 13. Field test to demonstrate the robot's driving performance. (a) Grass field. (b) Sand field with some stones. (c) Low space. (d) Gravel field. (e) Stream. Refer to multimedia extension for the experiment.

The size of the gravels ranged from 50 to 80 mm in width and 20 to 40 mm in height. Through this experiment, it was confirmed that driving is possible even on very irregular and rough terrain. Fig. 13(e) shows a driving test on a flowing stream. As a result of the experiment, it was confirmed that it was possible to drive in an environment with real water and that it was well waterproofed.

V. Conclusion

This article developed an amphibious robot that can drive on a water surface while maintaining the advantages of ASW. Therefore, an ASPW, an amphibious mechanism that did not exist before, is designed to achieve this purpose and create a robot that complements the shortcomings of previous amphibious robots. The paddle design was optimized using the Taguchi method. As a result of the experimental performance of the robot, the robot showed good performance in ground driving, overcoming obstacles, and climbing up in transition environment. The ground driving ability and obstacle overcoming ability depends on the length of the paddle, and the water surface driving ability depends on the area of the paddle, so if the size of the robot and paddle is increased, it will exhibit a faster speed and better ability to overcome obstacles. In the case of uneven terrain, such as an outdoor environment, it was observed that the robot could drive well. Although the robot's water surface driving speed is relatively small, future work will try to solve this by increasing the paddle size and designing the robot body to reduce fluid resistance.

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图：原文第 8 页，五类实地环境验证。

如何理解这篇论文的贡献

1. 贡献是“被动机械调制”，而不是增加一个桨叶舵机。桨叶姿态由轮内齿轮自动约束，控制系统仍可保持简单。
2. 作者用“普通桨轮完全浸没不前进”作为对照，直接验证了自转桨叶的必要性。

3. 地面速度模型与实验高度吻合，越障理论值也接近实测，说明机构的几何部分可预测；水动力部分则主要依赖实验优化。

局限与批判性阅读

- 水面最高速度仅 0.10 m·s⁻¹，适合低速侦察，不足以证明其在强流、浪涌或风浪条件下的救援能力。
- 论文的水上性能主要以直线速度表征，缺少转向半径、侧风/横流稳定性、能耗和续航比较。
- 过渡坡度结果受浮力、重心和轮地摩擦共同影响，但样机尺寸、载荷变化对极限坡度的敏感性没有系统展开。
- 齿轮和万向节数量较多，水下长期运行时的泥沙侵入、腐蚀、密封、磨损与维护成本仍需耐久试验。

可复用的设计启发

- 对周期运动机构，可优先思考“让有效面积/力臂随相位被动变化”，而不是直接增加执行器。
- 对机理难建模但试制成本低的对象，田口法适合先筛选主效应；后续可再用响应面或贝叶斯优化做精细搜索。
- 下一步实验应加入单位距离能耗、静水/流场下的姿态稳定性、转弯性能、密封寿命和载荷敏感性，才能从概念样机走向救援装备。

来源与使用说明

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